svyspline

`svyspline` is an R package for fitting and visualizing spline associations between continuous exposures and survival outcomes in complex survey data.

The main function, `fit_svyspline()`, fits survey-weighted Cox models with natural cubic splines and returns predicted hazard ratios (HRs), 95% confidence intervals, and a test for non-linearity.

Getting started

Install from GitHub:

```
install.packages("remotes")
remotes::install_github("MEB-PASS/svyspline")
```

Load the package:

```
library(svyspline)
```

Basic use

`fit_svyspline()` requires an existing `survey` design object:

```
res <- fit_svyspline(
  design = survey_design,
  exposure_var = "bmi",
  time_var = "fu_months",
  event_var = "mortality",
  covariates = c("age", "factor(sex)")
)
```

The function can optionally:

- trim extreme exposure values using survey-weighted quantiles
- place spline knots using survey-weighted quantiles
- define the HR reference value using a survey-weighted quantile
- generate predictions across the exposure range
- compare the spline association with a linear model
- test for non-linearity

Main arguments

Argument	Description	Default
`design`	Existing `survey` design object	-
`exposure_var`	Continuous exposure	-
`time_var`	Follow-up time	-
`event_var`	Event indicator	-
`covariates`	Covariates in the Cox model	`NULL`
`trim_probs`	Lower and upper quantiles used to trim the exposure	`c(0.01, 0.99)`
`knot_probs`	Quantiles used for spline knots and boundaries	`c(0.05, 0.50, 0.95)`
`ref_prob`	Quantile used for the HR = 1 reference	`0.10`
`grid_by`	Prediction grid spacing	`1`
`round_to`	Rounding unit for the reference and grid limits	`1`

For example:

```
res <- fit_svyspline(
  design = survey_design,
  exposure_var = "bmi",
  time_var = "fu_months",
  event_var = "mortality",
  covariates = c("age", "factor(sex)"),
  trim_probs = NULL,
  knot_probs = c(0.05, 0.50, 0.95),
  ref_prob = 0.10
)
```

Set `trim_probs = NULL` to use the full observed exposure range.

Returned results

The fitted object includes:

- `fit_spline` - survey-weighted Cox model with spline terms
- `fit_linear` - corresponding linear Cox model
- `pred_spline` - predicted HRs and 95% CIs from the spline model
- `pred_linear` - predicted HRs and 95% CIs from the linear model
- `knots` - spline knot locations
- `ref_val` - exposure value corresponding to HR = 1

- `nonlinearity_p` - p-value for the test of non-linearity
- `nonlinearity_p_label` - formatted non-linearity p-value
- `design_clipped` - survey design used for model fitting

For example:

```
res$knots
res$ref_val
res$nonlinearity_p_label
head(res$pred_spline)
```

Survey design and subsetting

Create the survey design and define survey weights before applying analytic or subgroup restrictions.

```
survey_design <- svydesign(
  data = dat,
  id = ~psu,
  strata = ~strata,
  weights = ~weight,
  nest = TRUE
)

survey_subset <- subset(
  survey_design,
  age >= 40
)
```

Pass the resulting survey design to `fit_svyspline()`.

Vignette

The package vignette provides a complete example of fitting and visualizing a survey-weighted spline model:

```
vignette("svyspline")
```

Reporting issues

Please report comments, suggestions, errors, and bugs through the GitHub Issues page.